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SMS group

Safety provisions for danger zones
Fence systems
Overview - Definitions

SN 401-1

ICS 13.110

Supersedes SN 401-1: 2010-03

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1 Scope

This company standard of SMS group specifies fence systems consisting of protective fences and access doors. The fence systems shall be used in equipment and for machinery of SMS Siemag every time persons have to be protected against hazards emanating from equipment/machines. This standard is intended to simplify and standardise the detail design of protective fences when such design is demanded.
Fence systems designed in compliance with the specifications of this standard fulfil the relevant European safety requirements which are applicable to equipment and machinery of SMS Siemag. For the use outside Europe the national regulations applicable in the respective countries shall be observed. Information on this can be obtained from the Senior Departmental Specialist for product safety.
Protective fences are fixed guards and doors are movable guards as defined by DIN EN 953. They are used to prevent access to the danger zone. The fences specified in this standard do not offer protection e.g. from ionising radiation, radiation heat or material/parts flying out such as strip residues, scale, etc. When such hazards exist, additional protective measures shall be taken which can be found in the respective project-related risk assessment.

2 Normative references

SN 401-2	Safety provisions for danger zones – Fence elements
SN 401-3	Safety provisions for danger zones - Door elements/door units
SN 401-4	Safety provisions for danger zones, electric door interlocking, design directive

3 General specifications

The specifications are based on hazards caused by moving machine parts or by material. If an equipment presents other hazards, it shall be checked for each particular case whether it is permitted to use the fence elements defined in this standard and, when necessary, the fence elements shall be replaced with suitable guards or such guards shall be used in addition to the fence elements.
The fence and door elements used shall be 1400 mm or 2200 mm high, intermediate heights are not permitted. Fence systems according to SN 401 can only be used at one level. When differences in level have to be provided with protective fences, the necessary elements shall be tailored for the particular application.
According to DIN EN ISO 13857, fences shall be installed at a defined minimum horizontal distance from a point of possible hazard located within the danger area behind the fence to prevent reaching of this hazard point (see Fig. 1). If it is not possible to comply with the minimum distances stated in Table 1, other solutions shall be implemented which are specifically tailored for the case of hazard taking into account the stipulations made in DIN EN ISO 13857 and DIN EN 349.

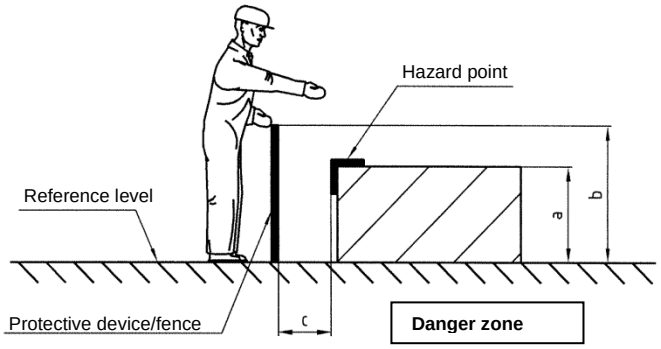
Table 1 - Dimensions

Height of hazard point ***a***	Height of protective device ***b***	
	1400	2200
	Horizontal minimum distance from hazard point ***c***	
- [a)]	≥ 1100	≥ 400
[a] Height of hazard point according to application		

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a = height of hazard point
b = height of protective device
c = horizontal minimum distance

Fig. 1 – Danger zone

4 Access to the danger zone, door safety categories

The fence system specified in the SN 401 set of standards uses different safety categories for access to the danger zone. On the basis of risk assessment according to SN 114 and with due regard to the requirements made in DIN EN 953, the safety categories applicable to the access doors shall be determined for access to the different equipment areas and danger zones. The different safety categories impose different safety devices which shall be installed without exception on every access door leading to the danger zone.
There are two safety categories for the access doors:

Safety category T I:
The safety device is purely mechanical (key, lock) and there is no connection with the control systems. Access to the protected area is possible for authorised persons (lockable doors). Warning signs according to SN 118 shall be provided in the door area.
Safety category T II:
The safety device is connected with the control system (safety door), different designs of the electric safety devices are possible. Access to the protected area is possible only after the equipment has been brought to a safe operating condition. The doors are secured by electric guard locking or key transfer system. The escape unlocking mechanisms of doors of safety category T II, height 1400 mm, shall be provided with a suitable guard to avoid actuation of the mechanism by reaching over the fence from outside. Type and construction of this guard are left to the discretion of the specialist company. Warning signs according to SN 118 shall be provided in the door area.

5 Fence and door elements, terms and definitions

A distinction is made between fence elements and door elements:

- Fence element is the general term for all fence nettings and fence posts (see Fig. 2). The fence elements stated shall be used for both safety categories. Fence posts can also be ordered as complete units with all necessary fastening elements.

Door element is the general term for all individual elements used to make up wing door units and their posts (see Fig. 2). The door elements have different designs for safety category T I (mechanical lock) and category T II (electric guard interlocking/key transfer system). All components required for category T II are specified in SN 401-4.

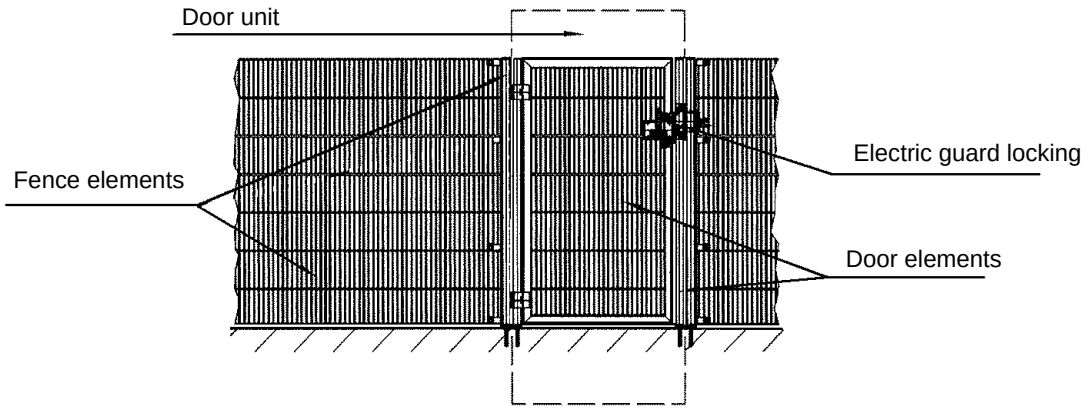


Fig. 2 – Individual elements

Referenced technical standards, codes and regulations

DIN EN 349	Minimum gaps to avoid crushing of parts of the human body
DIN EN 953	Guards; General requirements for the design and construction of fixed and movable guards
DIN EN ISO 13857	Safety of machinery; Safety distances to prevent hazard zones being reached by upper and lower limbs
SN 114	Plant safety; Plant-related risk assessment; Risk graph
SN 118	Safety layout; symbols

Revision of August 2011

New wording in the section on Scope.
Use of SMS group layout.